MIL-C-47269(MI)
9 August 1974
SUPERSEDING
MPD-2001A
24 April 1958

MILITARY SPECIFICATION

CASTINGS, INSPECTION OF
(FOR GUIDED MISSILE AND ROCKET MATERIEL)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes standards of soundness and freedom from defects for metal castings used in guided missile and rocket materiel and prescribes the requirements for inspection of castings to determine their conformance to such standards.

1.2 Classification. The classification of castings for purposes of inspection hereunder shall be as specified on the design drawing for the castings in accordance with the classifications established in MIL-C-6021.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-6021	Castings, Classification and Inspection of (for Aeronautical Applications)
MIL-I-6866	Inspection: Penetrant Method of
MIL-I-6868	Inspection Process; Magnetic Particle

FSC 1420

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Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society For Testing Materials (see 6.5)

ASTM E 98	Reference Radiographs for Inspection of Aluminum and Magnesium Castings

(Application for copies of the ASTM Standards should be addressed to American Society for TEsting Materials, 1916 Race Street, Philadelphia, PA, 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Basic requirement. Inspection of castings shall be in accordance with the requirements of MIL-C-6021 for the applicable class and as required herein.

3.2 Standards. Standards of soundness and freedom from defects which apply for the various areas of a casting shall be as specified on the drawing. (See 6.2 and 6.3). These standards shall be used to determine acceptability in accordance with Tables I, II and III of this specification. When no standard is specified on the drawing, Standard II hereof shall apply. When no standard is specified but the drawing does designate the "critical" areas of a casting, Standard II shall apply for the areas so designated, and Standard III shall apply for the remainder of the casting.

3.3 Visual inspection.

3.3.1 Gas holes and dross. Maximum acceptable sizes and spacing of visually detected gas holes and dross in the surface of a finished casting shall be as established in Table I. Except for Class IIB castings reworked as specified below, defects in excess of those specified in Table I shall be cause for rejection.

3.3.2 Reworking of Class IIB castings. Class IIB castings having exposed gas holes and dross in excess of those specified in Table I may be reworked by filling in the objectionable defects with weld material having

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substantially the same characteristics as the base metal. Such welded areas shall be so marked to be readily identifiable during and after subsequent finishing operations performed on the castings such as heat treating and sand blasting. In addition, each reworked casting shall be stamped in a designated area with the following marking: Three concentric circles, the largest diameter being 1/4 inch, together with the welder's identification made within or immediately adjacent to the circles. Radiographic inspection of reworked castings shall not be required.

3.4 Magnetic and penetrant inspection. All castings of ferromagnetic materials shall be inspected in accordance with MIL-I-6368. All castings of nonferromagnetic materials shall be penetrant inspected in accordance with MIL-I-6866. Penetrant inspection of aluminum alloy castings which are to be protected by anodic or other chemical film shall be performed before such anodic or chemical process, and shall not be required after the parts are so treated. For castings of all calsses, magnetic and penetrant inspection shall be for the purpose of detecting and evaluating cracks and cold shuts only. For Class 1A and Class IIA castings, the disposition of other defects shall be as determined by the results of radiographic inspection. For all classes of castings, gas holes and dross exposed on the surface are acceptable if within the limits established in Table I.

3.5 Radiographic inspection. Radiographs of casting shall be prepared in accordance with the requirements of MIL-C-6021 and any applicable radiographic positioning chart. Radiographs of aluminum alloy and magnesium alloy castings, whom such are required, shall be compared with the applicable reference radiograph of ASTM E98 as established in Tables II and III respectively, (see 6.4) Any defect in excess of those allowed in Table II or Table III for the particular standard shall be cause for rejection, except that such defect may be accepted in a raw casting if subsequent machining will completely remove the defect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements:

5. PREPARATION FOR DELIVERY

This section not applicable to this specification.

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6. NOTES

6.1 Intended use. This specification is intended for use when specifically called out on the design drawing or radiographic positioning chart for the castings involved.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

6.3 Definitions.

6.3.1 Drawing. The term "drawing" is used throughout this specification to mean the design drawing for the casting involved or the radiographic positioning chart, whichever is applicable.

6.3.2 Standard I. Standard I is primarily intended for highly stressed areas or pressure tightness. Special process control is usually required to meet this standard.

6.3.3 Standard II. Standard II is primarily intended for parts subject to medium stress where better than average quality is required.

6.3.4 Standard III. Standard III is primarily intended for other parts subject to low stressed areas of other castings.

6.3.5 Reference radiographs. Radiographs portrayed in ASTM E98 are identical to those in NAVAER-00-15PC-504, Navy Department, Bureau of Aeronautics, Aeronautical Inspection Manual, Vol. 3, Section 5, "Reference Radiographs for Inspection of Aluminum and Magnesium Castings," 1 August 1951. The manual may be used in lieu of ASTM E98.

6.6 Supersession data. This specification includes the requirements of Missile Purchase Description, MPD-2001A, dated 24 April 1958.

Custodian:
ARMY-MI

Preparing Activity:
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Table I Acceptability limits for size and spacing of gas holes and dross on surfaces 1/ of castings

Applicable standard specified on drawing	Maximum size 2/ of defect for castings classified under MIL-C-6021			Minimum spacing, edge-to-edge, between defects 3/	Minimum edge distance 3/
	Class 1A	Class IIA	Class IIB		
I	0	0	---	---	---
II	0	1/32 inch	---	10 D	2-1/2 D
III	1/32 inch	1/16 inch	---	8 D	2 D
none	1/16 inch	3/32 inch	3/32 inch	6 D	1-1/2 D

Notes:

1/ Not applicable to surfaces required to be continuous for sealing purposes (e.g. "O" ring groove, etc.); not applicable to threads; not applicable to fillets or corners within the edge distance listed in last column; not applicable to surface whose width is less than 5 times the maximum allowable size of defect.

2/ Major diameter or depth of defect, whichever is greater, Notation "0" means no defect is allowed which cannot be removed within drawing dimensional tolerances. Notation --- means no limit is established herein.

3/ D is the major diameter of the larger of two neighboring defects when determining spacing or of each single defect when determining edge distance. Notation --- means no limit is established herein.

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Table II Maximum acceptable defects in aluminum alloy casting

Type of defect	Reference radiograph set in ASTM E98-53T	Maximum allowable defect, by radiograph number in reference set or as otherwise noted		
		Standard I	Standard II	Standard III
Gas holes	1.1	2	4	6
Gas Porosity (round)	1.21	1	2	5
Gas Porosity (elong.)	1.22	1	4	5
Shrinkage cavity	2.1	1	3	4
Shrinkage Porosity or Sponge	2.2	1	3	5
Foreign Material (less dense material)	3.11	1	3	5
Foreign material (more dense material)	3.12	none	2	2
Segregation	3.2	none	1	1
Hot cracks	4.1	none	none	none
Cold cracks	(4.2)	dwg. tol.	dwg. tol.	dwg. tol.
Cold Shut	(4.3)	dwg. tol.	dwg. tol.	dwg. tol.
Surface irregularities	(5.1)	dwg. tol.	dwg. tol.	dwg. tol.
Misruns	(5.2)	dwg. tol.	dwg. tol.	dwg. tol.
Core Shift	(5.3)	dwg. tol.	dwg. tol.	dwg. tol.

NOTES:

(1) Notation "none" means no defect of the types illustrated in the reference radiograph set is allowed Notation "dwg. tol." means that defects of the types illustrated are allowed provided the casting, when machined, conforms to the desgin drawing and its established tolerances.

(2) Defects in excess of those allowed in this table shall not be cause for rejection if subsequent machining will completely remove them.

(3) Defects allowed by this table shall be cause for rejection when a combination of several is determined to be as harmful to mechanical properties as any single rejectable one.

(4) Subsurface gas holes or foreign material allowed by this table shall be cause for rejection when closer than twice maximum dimension to an edge or extremity of the finished part.

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Table III Maximum allowable defects in magnesium alloy casting

Type of defect	Reference radiograph set in ASTM E98	Maximum allowable defect, by radiograph number in reference set or as otherwise noted		
		Standard I	Standard II	Standard III
Gas Holes	1.1	1	2	3
Gas Porosity	1.2	none	1	2
Shrink Cavity	2.1	none	1	2
Microshrinkage	2.3	1	4	6
Foreign material (less dense material)	3.11	2	4	4
Foreign material (more dense material)	3.12	1	2	2
Hot Cracks	4.1	none	none	none
Cold Cracks	(4.2)	dwg. tol.	dwg. tol.	dwg. tol.
Cold Shuts	(4.3)	dwg. tol.	dwg. tol.	dwg. tol.
Surface irregularities	(5.1)	dwg. tol.	dwg. tol.	dwg. tol.
Misrun	(5.2)	dwg. tol.	dwg. tol.	dwg. tol.
Core Shift	(5.3)	dwg. tol.	dwg. tol.	dwg. tol.

NOTES:

(1) Notation "none" means no defect of the types illustrated in the reference radiograph set is allowed. Notation "dwg. tol." means that defects of the types illustrated are allowed provided the casting, when machined conforms to the design drawing and its established tolerances.

(2) Defects in excess of those allowed in this table shall not be cause for rejection if subsequent machining will completely remove them.

(3) Defects allowed by this table shall be cause for rejection when a combination of several is determined to be as harmful to mechanical properties as any single rejectable one.

(4) Subsurface gas holes or foreign material allowed by this table shall be cause for rejection when closer than twice their maximum dimension to an edge or extremity of the finished part.

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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